

Raj Shekhor Roy

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Professional Profile:    

Research Interest

Protein structure, interaction and quality prediction, novel protein discovery, protein and antigen design, and cryo-EM.

Education

University of Missouri-Columbia,

2019 -2024

Ph.D. in Computer Science

Missouri , USA

Thesis: Machine learning prediction of interchain contacts and structural model quality for protein complexes

University of Missouri-Columbia,

2019-2023

MS in Computer Science (awarded en route to Ph.D.)

Missouri , USA

BRAC University,

B.Sc. in Computer Science and Engineering

2013- 2017

Thesis: Human fall control system for disabled people using double inverted pendulum

Dhaka, Bangladesh

Academic Professional Experience

SPHERICAL

January 2025- Present

Postdoctoral Researcher, Texas A&M University

Antigen Pipeline

- Led the development and optimization of computational pipeline for antigen identification and characterization using bioinformatics and deep learning. The initial prototype demonstrated promising performance in detecting SARS-Cov-2 conserved epitopes. (publication in progress)
- Integrated structural, sequence analysis, and machine learning techniques improve antigen prediction and analysis.
- Collaborating with interdisciplinary teams to enhance pipeline performance and validate computational predictions with experimental data.
- Managed task allocation and project organization for graduate students, while fostering knowledge exchange and collaborative learning.

Agentic Biosynthetic Gene Cluster (BGC) Pathway Analysis

- Performed genome-based BGC annotation and gene analysis to identify biosynthetic proteins.
- Developed a pathway inference pipeline to connect clusters to predicted metabolic pathways and secondary metabolites.

City Sim Ai Lab

July 2024 - December 2024

Senior Graduate Research Assistant, University of Missouri

Columbia, Missouri

- Led the development of deep learning tools, encompassing data collection, preprocessing, and augmentation to enhance image dataset quality for model training.
- Developed a pipeline for object detection and segmentation of urban images.
- Conducted model training using state-of-the-art deep learning frameworks, optimizing performance through hyperparameter tuning and architectural adjustments.

August 2019 - July 2024

Bioinformatics and Machine Learning Lab

Columbia, Missouri

Graduate Research Assistant, University of Missouri

Protein Structure Reconstruction

- Developed DRCon, a dilated Resnet model that utilized distributed deep learning for the interchain contact prediction of homodimers. In addition, I created a dataset for training purposes from scratch. The tool surpassed the **state-of-the-art model's** performance at the time of release.
- Developed ConComplex, a simulated annealing based tool to reconstruct protein structure.
- Co-developed DeepComplex, a web server to predict protein structure.
- Co-developed DRLcomplex, a novel approach to reconstruct protein dimeric structure utilizing reinforcement learning.
- Contributed to GDFold, a gradient descent based method to reconstruct protein structure.

Protein Quality Assessment

- Developed MULTICOM_qa, a tool to estimate the global fold quality of a protein using a hybrid approach.
 - The average top-1 TM-score of the MULTICOM_qa for The Critical Assessment of protein Structure Prediction, CASP15 assembly targets is **5.3% higher than that of the standard AlphaFold-Multimer**.
 - It was **ranked top** in estimating the global fold accuracy of protein assembly structures in the 2022 CASP15 experiment.
 - Received multiple recognition for CASP15 contributions with a feature article by the Mizzou Engineering team and the Daily Star.

Cryo-EM

- Proficient in utilizing advanced computational tools such as CryoSPARC and EMAN2 for Cryo-EM studies.
- Experienced in reconstructing electron density maps from 2D micrograph images and enhancing these maps by refinement.
- Skilled in processing electron density maps and conducting qualitative assessments to ensure data integrity and reliability.

Honors and Patents

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| ● Outstanding PhD Student Award by the EECS department of the Missouri-Columbia. | 2024 |
| ● Outstanding PhD Student Award by the College of Engineering of the University of Missouri-Columbia. | 2024 |
| ● Lead the team that ranked no. 1 in estimating global fold accuracy of protein quaternary structures in CASP15 (EMA) | 2022 |
| ● Contributed to the 3rd ranked team among server predictors for tertiary or quaternary structure prediction in CASP15. | 2022 |
| ● DEEP LEARNING SYSTEMS AND METHODS FOR PREDICTING STRUCTURAL ASPECTS OF PROTEIN-RELATED COMPLEXES (Patent Filed) | 2022 |

Mentorship and Teaching Experience

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| ● Raiyan Chowdhury, PhD student at Texas A&M University | 2025-Present |
| ● Weifeng Zhang, PhD student at Texas A&M University | 2025-Present |
| ● ANM Nafiz Abeer, PhD student at Texas A&M University | 2025-Present |
| ● Lianhao Zhao, PhD student at Texas A&M University | 2025-Present |
| ● Nora Shih, PhD student at the University of Missouri- Columbia (co-advising with Dr. Jayedi Aman) | 2025-Present |
| ● Chloe Harmon, Under-grad student at the University of Missouri- Columbia (co-advising with Dr. Jayedi Aman) | 2025-Present |
| ● Teaching assistant for “Python 1: Learn to Program in Python” (INFOTC 4401) at the University of Missouri- Columbia | Fall 2024 |

Service and Leadership

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| ● Judge, AggiE_Challenge Expo, Texas A&M University | 2025-Present |
| ● Community Reviewer, Frontiers in Bioinformatics | 2025-Present |
| ● Organizer, Inter-lab Journal Club, Texas A&M University | |
| ● Reviewer, Frontiers in Molecular Biosciences | 2025-Present |
| ● Reviewer, Computational and Structural Biotechnology Journal | 2023-Present |
| ● Lead contributor, CASP15 protein quaternary structure quality assessment (EMA) | 2022 |
| ● Organizer, BML Lab Journal Club, University of Missouri. | 2021-2024 |
| ● Lead contributor, The Critical Assessment of protein Structure Prediction and Critical Assessment of PRedicted Interactions (CASP14-CAPRI50) | 2020 |
| ● Member, IEEE BRAC University Student Branch | 2017-2018 |
| ● Executive member, BRACU Computer Club | 2013-2017 |

Selected Talks

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| ● Talk, The 3rd Workshop on Advances in Systems Immunology (ASI) | 2025 |
| ● Talk, EECS Graduate Student Seminar, University of Missouri-Columbia | 2024 |
| ● Invited talk, Strauch Lab, Infectious Diseases Division, Washington University in St. Louis | 2024 |

- Talk, The International Conference on Intelligent Biology and Medicine (ICIBM). 2023
- Poster, The ARPA-E Energy Innovation Summit. 2022
- Poster, The CASP14 Virtual Symposium. 2020
- Talk, The 4th International Conference on Electrical Engineering and Information & Communication Technology (iCEEiCT). 2018

Peer-Reviewed Publications and Conferences

Raj S Roy, Jaemin Oh, A N M Nafiz Abeer, Maria I Giraldo, Tesuro Ikegami, Scott C Weaver, Nikolaos Vasilakis, Byung-Jun Yoon, Xiaoning Qian (2026). An integrated computational antigen discovery pipeline with hierarchical filtering for emerging viral variants. *NAR Molecular Medicine*, Volume 3, Issue 2, April 2026, ugag024, <https://doi.org/10.1093/narmme/ugag024>

Roy, R. S., Liu, J., Giri, N., Guo, Z., & Cheng, J. (2023). Combining pairwise structural similarity and deep learning interface contact prediction to estimate protein complex model accuracy in CASP15. *Proteins: Structure, Function, and Bioinformatics*, 91(12), 1889–1902. <https://doi.org/10.1002/prot.26542>

Lensink, M. F., Brysbaert, G., Raouraoua, N., Bates, P. A., Giulini, M., Honorato, R. V., van Noort, C., Teixeira, J. M. C., Bonvin, A. M. J. J., Kong, R., Shi, H., Lu, X., Chang, S., Liu, J., Guo, Z., Chen, X., Morehead, A., **Roy, R. S.**, Wu, T., ... Wodak, S. J. (2023). Impact of AlphaFold on structure prediction of protein complexes: The CASP15-CAPRI experiment. *Proteins: Structure, Function, and Bioinformatics*, 91(12), 1658–1683. <https://doi.org/10.1002/prot.26609>

Liu, J., Guo, Z., Wu, T., **Roy, R. S.**, Quadir, F., Chen, C., & Cheng, J. (2023). Enhancing alphafold-multimer-based protein complex structure prediction with MULTICOM in CASP15. *Communications Biology*, 6(1), 1140. <https://doi.org/10.1038/s42003-023-05525-3>

Liu, J., Guo, Z., Wu, T., **Roy, R. S.**, Chen, C., & Cheng, J. (2023). Improving AlphaFold2-based protein tertiary structure prediction with MULTICOM in CASP15. *Communications Chemistry*, 6(1), 188. <https://doi.org/10.1038/s42004-023-00991-6>

Giri, N., **Roy, R. S.**, & Cheng, J. (2023). Deep learning for reconstructing protein structures from cryo-EM density maps: Recent advances and future directions. *Current Opinion in Structural Biology*, 79, 102536. <https://doi.org/10.1016/j.sbi.2023.102536>

Soltanikazemi, E.*, **Roy, R. S.***, Quadir, F., & Cheng, J. (2022). A deep reinforcement learning approach to reconstructing quaternary structures of protein dimers through self-learning (p. 2022.04.17.488609). *bioRxiv*. <https://doi.org/10.1101/2022.04.17.488609>

Roy, R. S., Quadir, F., Soltanikazemi, E., & Cheng, J. (2022). A deep dilated convolutional residual network for predicting interchain contacts of protein homodimers. *Bioinformatics*, 38(7), 1904–1910. <https://doi.org/10.1093/bioinformatics/btac063>

Soltanikazemi, E., Quadir, F., **Roy, R. S.**, Guo, Z., & Cheng, J. (2022). Distance-based reconstruction of protein quaternary structures from inter-chain contacts. *Proteins*, 90(3), 720–731. <https://doi.org/10.1002/prot.26269>

Lensink, M. F., Brysbaert, G., Mauri, T., Nadzirin, N., Velankar, S., Chaleil, R. A. G., Clarence, T., Bates, P. A., Kong, R., Liu, B., Yang, G., Liu, M., Shi, H., Lu, X., Chang, S., **Roy, R. S.**, Quadir, F., Liu, J., Cheng, J., ... Wodak, S. J. (2021). Prediction of protein assemblies, the next frontier: The CASP14-CAPRI experiment. *Proteins: Structure, Function, and Bioinformatics*, 89(12), 1800–1823. <https://doi.org/10.1002/prot.26222>

Gao, M., Lund-Andersen, P., Morehead, A., Mahmud, S., Chen, C., Chen, X., Giri, N., **Roy, R. S.**, Quadir, F., Effler, T. C., Prout, R., Abraham, S., Elwasif, W., Haas, N. Q., Skolnick, J., Cheng, J., & Sedova, A. (2021). High-Performance Deep Learning Toolbox for Genome-Scale Prediction of Protein Structure and Function. 2021 IEEE/ACM Workshop on Machine Learning in High Performance Computing Environments (MLHPC), 46–57. <https://doi.org/10.1109/MLHPC54614.2021.00010>

Quadir, F.*, **Roy, R. S.***, Soltanikazemi, E.*, & Cheng, J. (2021). DeepComplex: A Web Server of Predicting Protein Complex Structures by Deep Learning Inter-chain Contact Prediction and Distance-Based Modelling. *Frontiers in Molecular Biosciences*, 8. <https://www.frontiersin.org/article/10.3389/fmolb.2021.716973>

Quadir, F., **Roy, R. S.**, Halfmann, R., & Cheng, J. (2021). DNCON2_Inter: Predicting interchain contacts for homodimeric and homomultimeric protein complexes using multiple sequence alignments of monomers and deep learning. *Scientific Reports*, 11(1), 12295. <https://doi.org/10.1038/s41598-021-91827-7>

Rashid, Md. R., Shuvra, N. K. D., **Roy, R. S.**, & Sarker, S. (2018). A Theoretical Approach of Fall Control System for Disable People Using Double Inverted Pendulum. 2018 4th International Conference on Electrical Engineering and Information Communication Technology (iCEEICT), 49–53. <https://doi.org/10.1109/CEEICT.2018.8628130>

Technical Skills

Deep learning (PyTorch, TensorFlow), structural bioinformatics (protein prediction, Cryo-EM, antigen pipelines, BGC annotation), software & databases (Python, Java, Django, PostgreSQL, Docker), high-performance computing (Slurm, containerized workflows).

Professional Experience

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February 2018 - June 2019

Programmer

Dhaka, Bangladesh

During my tenure, I amassed extensive experience and honed a versatile skill set, adept at implementing solutions across diverse programming languages to meet stringent deadlines while ensuring code version control.

- **Ekipay**

Ekipay, launched under the ICT Division of Bangladesh, facilitates the payment of utility bills and government fees using multiple online payment methods, including internet banking, mobile wallets, and debit/credit cards.

- Developed and maintained APIs for seamless integration with various payment systems.
- Designed scalable databases for efficient data storage and retrieval.
- Implemented automated data backup solutions to ensure data security.
- Collaborated with cross-functional teams to gather requirements and develop technical solutions.

- **G-ERP**

Government ERP tailored to meet the specific needs of Bangladesh Government entities, initially focusing on automating ICT functionalities and planning divisions, thereby enhancing operational efficiency and enabling a sharper focus on delivering optimal outcomes for stakeholders.

- Developed APIs to streamline system integration for a government ERP system.
- **Led** responsive frontend development to enhance user experience.
- Automated report generation for efficient data analysis.